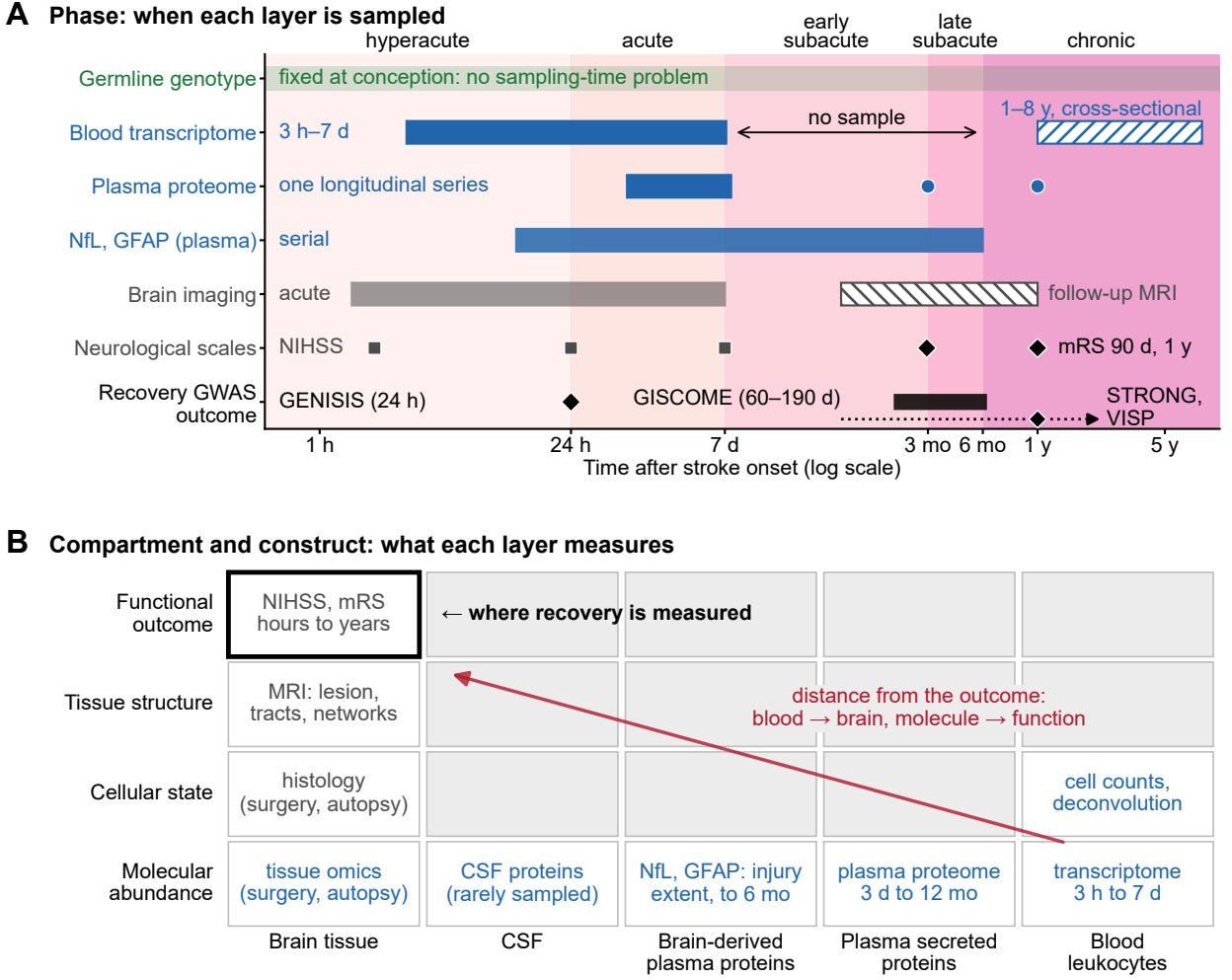

A
Phase: when each layer is sampled
hyperacute
acute
early subacute
late subacute
chronic
Germline genotype
fixed at conception: no sampling-time problem
Blood transcriptome
3 h–7 d
no sample
1–8 y, cross-sectional
Plasma proteome
one longitudinal series
NfL, GFAP (plasma)
serial
Brain imaging
acute
follow-up MRI
Neurological scales
NIHSS
mRS 90 d, 1 y
Recovery GWAS outcome
GENISIS (24 h)
GISCOME (60–190 d)
STRONG, VISP
1 h
24 h
7 d
3 mo
6 mo
1 y
5 y
Time after stroke onset (log scale)
B
Compartment and construct: what each layer measures
Functional outcome
NIHSS, mRS hours to years
← where recovery is measured
Tissue structure
MRI: lesion, tracts, networks
distance from the outcome:
blood → brain, molecule → function
Cellular state
histology (surgery, autopsy)
cell counts, deconvolution
Molecular abundance
tissue omics (surgery, autopsy)
CSF proteins (rarely sampled)
NfL, GFAP: injury extent, to 6 mo
plasma proteome 3 d to 12 mo
transcriptome 3 h to 7 d
Brain tissue
CSF
Brain-derived plasma proteins
Plasma secreted proteins
Blood leukocytes